AI Learning Roadmap (Step by Step)

🔥 Phase 1: Prerequisites (1-2 Months)

1️⃣ Learn Python

- **Topics to Learn:** Loops, functions, OOP, NumPy, Pandas
- **Resources:**
  - Automate the Boring Stuff with Python (Book)
  - Python for Everybody (Coursera)
  - Google's Python Class

2️⃣ Math for AI

- **Topics to Learn:**
  - **Linear Algebra:** Matrices, vectors, dot product, eigenvalues
  - **Calculus:** Differentiation, gradients, partial derivatives
  - **Probability & Statistics:** Bayes theorem, probability distributions
- **Resources:**
  - Mathematics for Machine Learning (Book)
  - Essence of Linear Algebra (YouTube - 3Blue1Brown)
  - Khan Academy - Calculus

🔥 Phase 2: Fundamentals of AI & Machine Learning (2-3 Months)

3️⃣ Introduction to Machine Learning

- **Topics to Learn:** Supervised & Unsupervised Learning, Regression, Classification, Clustering
- **Resources:**
  - Machine Learning by Andrew Ng (Coursera)
  - Hands-on Machine Learning with Scikit-Learn & TensorFlow (Book)
  - StatQuest - Machine Learning Explained

4️⃣ Programming Machine Learning Models

- **Topics to Learn:** Train-test splits, feature engineering, overfitting, hyperparameter tuning
- **Resources:**
  - Python Machine Learning by Sebastian Raschka
  - Fast.ai - Practical Deep Learning

5️⃣ Deep Learning Basics

- **Topics to Learn:** Neural Networks, Backpropagation, Activation functions, Loss functions
- **Resources:**
  - Deep Learning Specialization (Andrew Ng)
  - Neural Networks from Scratch (YouTube)

🔥 Phase 3: Advanced AI Topics (3-5 Months)

6️⃣ Natural Language Processing (NLP)

- **Topics to Learn:** Tokenization, Word Embeddings, Transformers, LLMs like GPT
- **Resources:**
  - Speech and Language Processing (Book)
  - Hugging Face Transformers Course

7️⃣ Computer Vision

- **Topics to Learn:** CNNs, Image Classification, Object Detection, GANs
- **Resources:**
  - CS231n: Deep Learning for Computer Vision (Stanford)
  - Fast.ai - Practical Deep Learning

8️⃣ Reinforcement Learning

- **Topics to Learn:** Markov Decision Process, Q-learning, Policy Gradients, DQN
- **Resources:**
  - Reinforcement Learning Specialization (Coursera)
  - Reinforcement Learning: An Introduction (Sutton & Barto)

🔥 Phase 4: Practical AI Projects (Ongoing)

- **Build real AI applications** like chatbots, image classifiers, stock price predictors
- **Resources for Projects:**
  - Machine Learning Projects for Beginners
  - Kaggle (Datasets & Competitions)

🔥 Phase 5: MLOps (After Completing AI Roadmap)

9️⃣ Introduction to MLOps (Machine Learning Operations)

- **Topics to Learn:**

    - Model Deployment (APIs, Cloud, Docker, Kubernetes)
    - Model Monitoring (Performance tracking, retraining models)
    - CI/CD for Machine Learning (Automating updates)
    - Version Control for ML Models
- **Resources:**
    - MLOps Specialization (Coursera)
    - Google Cloud MLOps Guide
    - Practical MLOps (Book)

🔥 Phase 6: Keep Learning & Specialize

- **Read AI Research Papers:**
    - ArXiv AI Papers
    - Google AI Blog
- **Follow AI Trends:**
    - YouTube Channels: Two Minute Papers, Yann LeCun
    - AI Newsletters: The Batch (DeepLearning.AI)
- **Pick a Specialization:** AI in **business, finance, robotics, economics**, etc.

Summary of Your Learning Path

✅ **Phase 1:** Python, Math
✅ **Phase 2:** Machine Learning & Deep Learning
✅ **Phase 3:** Advanced AI (NLP, Computer Vision, RL)
✅ **Phase 4:** Hands-on Projects
✅ **Phase 5:** MLOps (Deploying & Scaling AI Models)
✅ **Phase 6:** Stay Updated & Choose a Specialization

This roadmap will take **6-12 months**, depending on your pace. You're on the right track! 🔥
